Sepehr Mahmoudian

Senior AI Developer

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github.com/sepehrmn
Berlin-Brandenburg

Nationality: German
Languages: German (Good), English (Fluent), Persian (Native)
Availability: Immediate

Over ten years of experience with neural networks, machine learning, and simulation of distributed systems, in both small and large teams. Proven track record in developing and implementing innovative solutions: from classical AI methods to modern multimodal agentic systems.

PROFESSIONAL EXPERIENCE

Hacker Room Resident

09/2025 – 01/2026

Merantix AI Campus • Berlin, Germany

- Developed Torus Automations Dreamweave product for generative media creation.
- Integrated Gaussian Splatting and Diffusion Models for precise temporal and spatial anchoring.
- Contracted with Near Foundation for Rust smart contracts on confidential computing and sovereign AI.
- Deployed on Cloudflare Workers with D1 (SQLite), R2, and additional edge features.
- Built robust full-stack architecture using Tanstack Start (Vite-based).
- Implemented multi-cloud approach (Cloudflare, GCP Cloud Run, AWS EC2).
- Actively contributed to AI events, including the Merantix Unconference.

Senior Artificial Intelligence Developer

07/2024 – 07/2025

Vision Labs UG (haftungsbeschränkt) • Berlin & Hamburg, Germany

- Conducted Proof-of-Concepts (PoCs) and developed applications in TypeScript and Python
- Developed and fine-tuned multimodal agent-based systems (NLP, Vision)
- Prompt engineering (DSPy), solution evaluation (LangFuse), and iterative improvement
- Designed, built, deployed, and tested data pipelines (Azure, GCP)
- Managed staging, testing, and production environments. Created unit tests and CI/CD pipelines.
- Consulted with clients on projects, feasibility, and methodologies

Research Associate / Software Developer

07/2021 – 07/2024

Freie Universität Berlin • Berlin, Germany

- Software development (C/C++, Python) for model scaling and optimization
- Applied RAG (Retrieval Augmented Generation) for technical literature analysis
- Evaluated reliability of agentic systems
- Conducted code and peer reviews in an international team
- Large-scale neural network simulations
- Statistical data analysis using ML techniques and Python

Research Associate / Software Developer

07/2019 – 06/2020

Georg-August-Universität Göttingen • Göttingen, Germany

- Research in neural networks and data analysis

Research Associate / Software Developer

03/2017 – 07/2019

Goethe-Universität Frankfurt • Frankfurt am Main, Germany

- Development of simulations and analysis tools for neuroscientific research

Visiting Scientist

03/2017 – 03/2019

Max Planck Institute for Dynamics and Self-Organization • Göttingen, Germany

- Interdisciplinary research on complex systems and neural networks

Research Associate / Software Developer

02/2015 – 03/2017

Forschungszentrum Jülich • Jülich, Germany

- Development of simulation software for neural networks and reinforcement learning

Software Developer

09/2010 – 03/2012

Centre for Content Creation • Cyberjaya, Malaysia

- Full-stack development and iPhone/iPad apps

EDUCATION

Dr. rer. nat. (PhD)

2020 – 2026 (exp.)

Technische Universität Darmstadt • Darmstadt, Germany

- Partial Information Decomposition for the design and understanding of neural networks and agentic systems.
- Automation of neural network simulations using agent systems

Master's by Research in Biosciences

2012 – 2014

University College London • London, United Kingdom

Computational Neuroscience • Graduated with Distinction (Top 10%, CGPA 4.0/4.0)

Bachelor of Science (Honours)

2009 – 2012

Anglia Ruskin University • Cambridge, United Kingdom

Graduated with First Class Honours (Top 1%, CGPA 4.0/4.0)

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Berlin, November 29, 2025